

URL Shortener — Project Overview

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: /home/dominic/Documents/fastapiurl

Report #: 1/20

Overview

FastAPI-based URL shortener with SQLite, click tracking, and multi-format API responses.

Purpose

Provides URL shortening with analytics and redirect tracking.

Status

Active development. See report 16 (Known Issues) and report 19 (Improvements).

Key Features

• URL shortening with custom aliases • Click tracking and analytics • JSON/Redirect/HTML responses • SQLite persistence

Entry Points

```
python main.py or uvicorn main:app
```

Detailed Architecture

Project Structure

```
app/main.py (FastAPI), app/routers/auth_router.py, app/routers/urls.py, app/models.py, app/auth
```

Technology Stack

Layer	Technology	Purpose
Backend Framework	FastAPI	REST/API server
Database	SQLite	Data persistence
Auth	JWT	Authentication & authorization
Frontend	React 19	User interface
Deployment	Render.com + GitHub Pages	Hosting & CI/CD

Module Breakdown

See project structure above.

Data Flow

1. Request enters via HTTP (REST, GraphQL, or WebSocket)
2. Authentication middleware validates JWT token (if required)
3. Router dispatches to appropriate handler
4. Handler processes business logic with database queries
5. Response formatted and returned to client
6. Frontend renders response in appropriate view

Entry Points

- **Server:** `uvicorn app.main:app`
- **Port:** 8000 (default)
- **Health check:** `GET /health`

Configuration

Environment Variables

Variable	Description	Default
DATABASE_URL	Database connection string	sqlite:///data.db
SECRET_KEY	JWT signing secret	(required)
CORS_ORIGINS	Allowed CORS origins	*
DEBUG	Debug mode flag	false

Deployment

Requirements

- Python 3.10+
- pip dependencies (see requirements.txt)
- SQLite (or PostgreSQL for production)

Startup Commands

```
# Development
uvicorn app.main:app --reload

# Production
uvicorn app.main:app --host 0.0.0.0 --port 8000
```

Related Reports

- Report 02: Architecture Analysis (detailed component diagram)
- Report 05: Tech Stack & Dependencies (version-specific)
- Report 13: Deployment Guide (Docker, CI/CD)